Pure Substances & Mixtures

OBJECTIVE

In this lesson, students will learn that all matter is classified as either a pure substance or a mixture.

ACARA CONTENT DESCRIPTIONS

Mixtures, including solutions, contain a combination of pure substances that can be separated using a range of techniques (ACSSU113)

- recognising the differences between pure substances and mixtures and identifying examples of each

LESSON PLAN

Activities	Resources
Activity 1: What Makes up Our Air? Open the presentation to the first slide and discuss the following question with students: • What makes up our air? Students might not be able to name all of the different gases that make up air, but most should be able to acknowledge that oxygen and carbon dioxide are present. Use the next two slides of the presentation to reiterate the fact that air is a mixture of gases and not a pure substance. 10	• Presentation: Pure Substances and Mixtures
Activity 2: Pure Substances vs. Mixtures Play Chapters 1 and 3 of the video and encourage students to take notes. By the end of the video, students should have developed an understanding of some of the features of pure substances and mixtures. Using the remaining slides of the presentation, discuss the key features of pure substances and mixtures and how their particles differ in terms of: - types of particles present - the two possible kinds of pure substances (elements and compounds) 15	• ClickView video *Types of Pure Substances, Solutions and Mixtures* Chapter 1: **https://clickv.ie./w/lp/6a** Chapter 3: **https://clickv.ie./w/lp/6b** • Presentation: Pure Substance and Mixtures
Activity 3: Classify Them! Give out the Classify Them! 1 and 2 worksheets and have students work in pairs to complete the cut-and-paste activity. Students can paste their work on a blank sheet of paper or in their notebooks. Once students have completed the worksheets, encourage them to share their classifications from the Classify Them! 2 worksheet. 25	• Photocopies of the Classify Them! 1 and 2 worksheets • Scissors, glue, blank sheets of paper/notebooks
Activity 4: Is It True or False? Give out the Is It True or False? worksheet to each student pair. Ask them to cut the worksheet in half. Have students do the worksheet individually to assess their understanding of the topics taught. 10	• Photocopies of the Is It True or False? worksheet

ANSWERS

Classify Them!

Example	Pure Substance		Mixture	If it is a mixture, what different components does it contain?
	Element	Compound		
Chocolate chip ice-cream			✓	cream, milk, sugar, chocolate flavouring, chocolate chips
Sugar		✓		
Vegemite			✓	yeast extract, salt, malt extract
Mercury	✓			
Air			✓	water vapour, nitrogen, oxygen, argon, carbon dioxide
Cake			✓	flour, eggs, milk, butter, sugar
Diamond	✓			
Water		✓		
Helium	✓			
Neon	✓			
Salt water			✓	salt, water
Baking soda		✓		
Gold	✓			
Salt		✓		
Sand			✓	silicon dioxide with other rocks and minerals

Is It True or False?

	Statement	True	False
1	Air is a mixture of several gases.	✓	
2	Water collected from the ocean is a pure substance.		✓
3	A pure substance contains more than one kind of particle.		✓
4	Vegetable soup is a mixture.	✓	
5	An element is a pure substance.	✓	
6	A compound is a mixture.		✓
7	When you combine two different pure substances, it is still a pure substance.		✓

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Classify Them! 1

Instructions:

1. Cut out Box A and paste it on the top of a blank sheet of paper or a page in your notebook.
2. Cut out each example in Box B.
3. Decide if each example is a pure substance or a mixture and paste it under the correct heading.
4. Complete the Classify Them! 2 Worksheet.

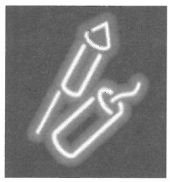
① Box A

MATTER

Pure Substance | **Mixture**

② Box B

cut along this line

Chocolate chip ice-cream 	Helium 	Baking soda 
Sugar 	Neon 	Air 
Vegemite 	Salt water 	Gold 
Mercury 	Diamond 	Salt 
Water 	Cake 	Sand 

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Classify Them! 2

Decide whether each example is a pure substance or a mixture. Classify them according to the table below. If it is a mixture, list its components.

Example	Pure Substance		Mixture	If it is a mixutre, what different components does it contain?
	Element	Compound		
Chocolate chip ice-cream				
Sugar				
Vegemite				
Mercury				
Air				
Cake				
Diamond				
Water				
Helium				
Neon				
Salt water				
Baking soda				
Gold				
Salt				
Sand				

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Is It True or False?

Indicate whether each statement is true or false.

	Statement	True	False
1	Air is a mixture of several gases.		
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